Pratham Aggarwal

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Aspiring researcher in LLM and diffusion model interpretability, with the goal of pursuing a PhD in machine learning after graduation.

Education

University of California San Diego — Expected Mar 2027
Bachelor of Science in Data Science; Mathematics-Computer Science
Coursework: CSE 151A (DSC 140B Equivalent), CSE 150A, DSC 148 — **Honors:** HDSI Scholar; HKN-IEEE Awardee

Skills

Programming Languages	Python, SQL, C, C++, Java
AI / Machine Learning	PyTorch, TensorFlow, Scikit-Learn, PowerBI, Tableau, Matplotlib, Pandas
Systems & Tools	Docker, Kubernetes, Git/GitHub, HPC (NCAR Derecho, NCAR Casper; NSF NRP), Google Cloud

Experience

Student Researcher @ Belkin Lab, Halıcıoğlu Data Science Institute — Apr 2026 – Present
- Researching **activation steering** for controllable refusal in **masked diffusion language models** (LLaDA-8B); extracted topic-specific and global refusal directions via single-pass CAA across all 32 layers.
- Used **layer-wise probing** to show topic identity is linearly decodable (**54% vs. 25% chance**); found topic directions are **geometrically distinct**, with self-harm near-orthogonal to global refusal.

Machine Learning Researcher @ Climate Analytics Lab, Scripps Inst. of Oceanography — Jun 2025 – Present
- Coupled a lightweight U-Net flux emulator with a physics-based ocean GCM (POP), accelerating ocean climate simulation by **20**× over state-of-the-art while preserving accuracy across a **40-year** autoregressive rollout.
- Recovered distributional extremes and fine spatial structure lost by regression baselines, generating **20–50 member ensembles** per timestep, by training a **conditional diffusion model** (EDM) on **155 years** of simulation data.
- Coupled the U-Net to POP via a custom file-handshake that bypasses the CESM coupler, enabling **128-rank** MPI ocean simulations on a global tripole grid on NCAR's Derecho supercomputer (test R^2 **0.62–0.98**).
- Extended a coupled ML–ocean simulation from **crashing in under 7 days to running for 500 years**, by diagnosing the numerical cause of the crashes and designing a fix that eased the model into stable operation.

AI/ML Software Developer Intern @ NIOV Labs Corporation — Jan 2026 – Apr 2026
- Built an NLP-based system to extract goals, tasks, participants, and deadlines from meeting transcripts, achieving **88%** agreement with human-labeled results across **20+** validation transcripts.
- Evaluated automated scheduling using extracted meeting details and calendar availability, cutting manual scheduling effort by **60%** while achieving 95% successful confirmations with 2-second response times.
- Containerized the application with **Docker** and deployed it on **Kubernetes**, improving reliability and enabling scalable deployments across environments.

Robotics Research Assistant @ Existential Robotics Lab — Aug 2025 – Dec 2025
- Benchmarked and optimized motion-planning algorithms (A*, RRT, MPC), achieving a **28%** reduction in path-planning failures and maintaining average trajectory error below **0.15m** in dynamic environments.
- Enhanced PyBullet-based simulation pipelines for mapping and localization by validating algorithm performance across **100+** experimental trials.
- Improved simulation speed by **2.1x** using physics evaluations and concurrent testing of **10+** montion planning algorithms.

Quantitative Researcher @ SFIC Quantitative Technologies — Mar 2025 – Jun 2025
- Improved simulated options trading returns by **12%** for **$1.3M** student-run investment fund by developing a Deep Learning agent with adaptive policy learning and early exercise strategies.
- Improved trading decisions by **18%** through a sequential decision-making step that dynamically selected option type, expiration, and exercise timing from historical time-series data.
- Reduced stock price prediction error to **0.96 MSE** by training an **LSTM** model on **5 years** of market data with feature engineering.

Data Science Intern @ Solana Center — Jan 2025 – Mar 2025
- Led a **5-member** data analytics team to process **2,000+** composting records using **Python** and **SQL**, developing a real-time **Power BI** dashboard that improved evaluation accuracy by **6.7%** and guided program decisions.
- Boosted dataset reliability by **9%** through automated data manipulation, cleaning and imputation routines, ensuring consistent and reproducible analytics across stakeholders.

Projects

NSF Hackathon (1st Place): Coastal Flooding Prediction | ***CatBoost, Feature Engineering*** — Jan 2026
- Built and evaluated ML models to predict coastal flooding risk during a CERN-hosted hackathon, selecting CatBoost as the best-performing approach and achieving a **0.94 F1 score** on the final benchmark.
- Improved flood prediction accuracy by comparing multiple models and tuning features on large-scale data, validating results through competition-based evaluation on CodaBench.

Hackfrontier (1st Place): Homeless Services Computer Vision Platform | ***Scikit-learn, Pandas, OpenCV*** — Jun 2025
- Built a geospatial forecasting platform that integrated **35+** transit, demographic, and geographic features, achieving **67%** accuracy in optimizing placement of homeless service centers.
- Deployed a real-time computer vision system with Oxen.ai and EyePop.ai to track transient homeless populations, delivering actionable insights for data-driven service allocation.